



# Efficacy and safety of tirofiban after successful endovascular reperfusion in acute ischaemic stroke (ATTRACTION) in China: a multicentre, double-blind, randomised controlled trial

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## Summary

**Background** Successful reperfusion after endovascular thrombectomy does not consistently result in functional independence in acute ischaemic stroke. We aimed to assess the efficacy and safety of tirofiban, a glycoprotein IIb/IIIa receptor antagonist, given to patients with acute ischaemic stroke who had had a successful endovascular reperfusion.

**Methods** This multicentre, double-blind, randomised controlled trial at 82 hospitals in China included patients with acute ischaemic stroke due to anterior-circulation large-vessel occlusion who had had a successful reperfusion after thrombectomy. Eligible patients were randomly assigned (1:1) to receive either tirofiban (intra-arterial bolus 5 µg/kg followed by intravenous infusion 0·1 µg/kg per min for 24 h) or placebo (administered with the same volume and according to the same bolus and infusion procedures as tirofiban), using computer-generated randomisation with fixed blocks stratified by study site. Patients, treating clinicians, investigators, and outcome assessors were masked to group assignments. The primary efficacy outcome was functional independence at 90 days (with a modified Rankin Scale score of 0–2), assessed in all randomly assigned participants (intention-to-treat population). Safety outcomes were symptomatic intracranial haemorrhage within 48 h, any evidence of intracranial haemorrhage on imaging within 48 h, and death within 90 days, and they were assessed in patients having received the study treatment with at least one safety evaluation. The Adjunct Tirofiban Treatment after Successful Endovascular Thrombectomy Recanalisation in Acute Anterior Circulation Ischaemic Stroke (ATTRACTION) trial is registered with ClinicalTrials.gov, NCT06265051 and is now completed.

**Findings** Of 1686 patients assessed, 1380 were randomly assigned to either the tirofiban group (689 patients) or the placebo group (691 patients) between April 9, 2024, and Sept 29, 2025. Median age was 71 years (IQR 62–77), 591 (43%) patients were female and 789 (57%) were male, and 1367 (99%) were of Han Chinese ethnicity. No patients were lost to follow-up at 90 days. Functional independence at 90 days was recorded in 340 (49%) of 689 patients in the tirofiban group and 299 (43%) of 691 patients in the placebo group (unadjusted absolute risk difference 6·1 percentage points, 95% CI 0·8–11·3,  $p=0\cdot023$ ; adjusted risk ratio 1·15, 95% CI 1·03–1·27,  $p=0\cdot0092$ ). There was no significant difference between study groups in the proportion of patients with symptomatic intracranial haemorrhage within 48 h (82 [12%] of 687 patients in the tirofiban group vs 65 [9%] of 691 patients in the placebo group), the proportion with any intracranial haemorrhage within 48 h (235 [34%] patients vs 219 [32%] patients), and 90-day mortality (126 [18%] patients vs 131 [19%] patients).

**Interpretation** In patients with acute ischaemic stroke due to anterior-circulation large-vessel occlusion achieving successful reperfusion, adjunctive tirofiban increased the likelihood of functional independence compared with placebo. Although symptomatic intracranial haemorrhage occurred numerically more often with tirofiban, the between-group difference was not significant, and this finding warrants caution when weighing potential benefit against bleeding risk.

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## Introduction

Endovascular thrombectomy (EVT) has improved outcomes in patients with acute ischaemic stroke due to large-vessel occlusion.<sup>1</sup> Nevertheless, substantial

disability persists in a considerable proportion of patients despite angiographic evidence of successful reperfusion.<sup>2</sup> The mechanisms underlying this clinical-angiographic dissociation are likely to be multifactorial and might

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See Online for appendix 1

See Online for appendix 2

## Research in context

### Evidence before this study

Although tirofiban has been widely used in China as an adjunct to endovascular thrombectomy, high-quality evidence supporting its efficacy and safety in this setting is insufficient. We searched PubMed from database inception to March 31, 2026, using the terms (tirofiban) AND (endovascular thrombectomy) AND ((successful recanalization) OR (successful reperfusion) OR (successful thrombectomy)), with no language restrictions. Although a pre-procedural randomised trial, RESCUE BT, evaluated tirofiban administered before thrombectomy, its findings did not directly address the effects of tirofiban after successful reperfusion. No randomised controlled trials specifically investigated the use of tirofiban after successful reperfusion following endovascular thrombectomy in patients with anterior-circulation large-vessel occlusion. Evidence for tirofiban initiated after endovascular thrombectomy in this patient population was available from multiple small, single-centre observational studies; however, the quality of this evidence was low and did not provide definitive guidance. These studies suggested that early post-procedural administration of low-dose tirofiban might improve functional outcomes without substantially increasing the risk of symptomatic intracranial haemorrhage. Overall, the current evidence was heterogeneous and insufficient to guide clinical practice.

### Added value of this study

The ATTRACTION trial is a large-scale, multicentre, double-blind, randomised controlled trial that evaluates the efficacy

and safety of adjunctive tirofiban in patients with acute ischaemic stroke due to anterior-circulation large-vessel occlusion who achieved successful reperfusion after endovascular thrombectomy. The trial findings showed that patients treated with tirofiban were more likely to achieve functional independence at 90 days than those who were given placebo. However, symptomatic intracranial haemorrhage was numerically more frequent in the tirofiban group than in the placebo group, although the overall difference was not significant. Exploratory subgroup analyses also raised a possible safety concern in patients with cardioembolic stroke, requiring cautious interpretation and further evaluation. The trial supports consideration of tirofiban as a promising post-reperfusion adjunctive strategy.

### Implications of all the available evidence

The ATTRACTION trial supports further consideration of tirofiban as a potentially useful adjunct after successful endovascular reperfusion, while highlighting the need for caution regarding bleeding risk. The trial also suggests the potential importance of early, targeted platelet inhibition in the immediate post-reperfusion period and invites continued investigation into early antiplatelet approaches after successful endovascular reperfusion aimed at improving microvascular reperfusion and neurological recovery.

include the extent of the baseline ischaemic core, distal embolisation, the no-reflow phenomenon, and re-occlusion.<sup>3</sup>

Tirofiban is a reversible glycoprotein IIb/IIIa inhibitor that inhibits the final common pathway of platelet aggregation.<sup>4</sup> It has been investigated as an adjunctive antiplatelet strategy in acute ischaemic stroke after intravenous thrombolysis.<sup>5</sup> Tirofiban was shown to be associated with less thrombus propagation and fewer downstream microthrombotic events,<sup>6</sup> which provides a rationale for its use during the immediate post-thrombectomy period when endothelial injury and platelet activation can be pronounced.<sup>7,8</sup> Evidence for tirofiban initiated after EVT remains scarce and is derived largely from small, single-centre observational studies.<sup>9–11</sup>

We performed the Adjunct Tirofiban Treatment after Successful Endovascular Thrombectomy Recanalisation in Acute Anterior Circulation Ischaemic Stroke (ATTRACTION) trial to evaluate the safety and efficacy of adjunct tirofiban initiated after successful reperfusion in patients with ischaemic stroke due to anterior-circulation large-vessel occlusion.

## Methods

### Study design

The ATTRACTION trial was an investigator-initiated, multicentre, double-blind, randomised, placebo-controlled trial at 82 hospitals in China. The trial was conducted in accordance with the principles of the Declaration of Helsinki and the Good Clinical Practice guidelines of the International Council for Harmonization. Ethics approval was granted by the Ethics Committees of the Tongji Medical College, Huazhong University of Science and Technology (2023S206) and all the participating centres. The trial protocol was published, and the protocol and statistical analysis plan are available in appendix 2.<sup>12</sup> The trial is registered with ClinicalTrials.gov (NCT06265051; registered on Feb 19, 2024) and is now completed. This Article was prepared in accordance with the CONSORT 2025 statement.

An independent data and safety monitoring board periodically reviewed accumulating safety data and trial conduct to ensure participant safety and trial integrity. An independent clinical event committee, unaware of trial-group assignments, adjudicated efficacy endpoints, safety endpoints, procedure-related complications, and

serious adverse events according to prespecified definitions. Neuroimaging data were reviewed centrally by an independent imaging core laboratory with readers who were unaware of treatment assignment. A standardised framework for data quality assurance and monitoring was implemented throughout the trial. This framework included automated checks in the electronic data capture system, central data-quality review, formal query resolution, and on-site and remote monitoring; details are provided in appendix 1 (pp 16–17).

### Participants

Patients were eligible if they were aged 18 years or older and had had an acute ischaemic stroke due to an anterior-circulation large-vessel occlusion within 24 h from the time they were last known to be well. Key inclusion criteria were a National Institutes of Health Stroke Scale (NIHSS) score of 6–30 immediately before trial enrolment, a pre-stroke modified Rankin Scale (mRS) score of 0 or 1, imaging-confirmed occlusion of the intracranial internal carotid artery or of the M1 or M2 segment of the middle cerebral artery, and an Alberta stroke program early CT score (ASPECTS) of 6 or higher on baseline non-contrast CT or diffusion-weighted MRI. Patients were eligible only if successful reperfusion (modified thrombolysis in cerebral infarction [mTICI] grade 2b or 3)<sup>13</sup> was documented either after completion of EVT or on diagnostic angiography showing spontaneous improvement of the occlusion to mTICI 2b–3, with no mechanical thrombectomy planned. Information about sex (male or female) and ethnicity was collected from identity cards. Written informed consent was obtained from each patient or, in cases of impaired decision-making capacity, from a legally authorised representative before randomisation.

Key exclusion criteria included intra-arterial thrombolysis after the index stroke, receipt of tirofiban within 24 h before EVT, bleeding within 30 days or surgery within 14 days before stroke onset, and planned use of antiplatelet therapy (including tirofiban) outside the trial regimen during the first 20 h after study therapy started. Accordingly, before randomisation, investigators confirmed that no periprocedural indication was anticipated for open-label antiplatelet therapy (including tirofiban); patients for whom such therapy was planned or expected were not enrolled. Additional imaging-based exclusion criteria included intracranial haemorrhage on CT or MRI, midline shift, cerebral herniation or mass effect with ventricular effacement, acute bilateral infarctions or multiple intracranial-vessel occlusions, and isolated external carotid artery occlusion. Full eligibility criteria are provided in appendix 1 (pp 9–10).

### Randomisation and masking

Patients were enrolled by site investigators who were masked to treatment allocation throughout the study and had no access to the randomisation sequence. After

eligibility was verified and written informed consent was provided, patients were randomly assigned (1:1 ratio) to receive either tirofiban or placebo through a centralised, web-based randomisation system with permuted blocks of four, stratified by study site. The allocation sequence was generated centrally by an independent statistical team using a computerised system. Patients, treating clinicians, investigators, and outcome assessors were unaware of trial-group assignments throughout the trial. Study medication kits were identical in appearance and labelled with a unique random number.

### Procedures

EVT was performed according to local practice with devices approved by the National Medical Products Administration and could include stent retriever, aspiration thrombectomy, or a combination of techniques. Intraprocedural rescue therapy, defined as angioplasty or stent placement, was permitted. Such rescue therapy was allowed only when the site investigator judged that the procedure could be completed without a periprocedural indication for open-label antiplatelet therapy (including tirofiban) outside the trial regimen, so that only the assigned study drug would be administered. Immediately after randomisation, the study drugs were administered according to protocol by physicians or nurses. In the tirofiban group, tirofiban was provided as a commercially available ready-to-use injection solution at a concentration of 0.05 mg/mL and was administered as an intra-arterial bolus dose of 5 µg/kg (maximum bolus dose, 0.5 mg), delivered through a catheter positioned in the artery proximal to the occlusion site, at 1 mL/min, followed by an intravenous infusion of 0.1 µg/kg per min continued for up to 24 h. The regimen was supported by previous clinical studies evaluating tirofiban administration routes in patients with acute ischaemic stroke with successful recanalisation and by expert consensus on the clinical application of tirofiban in China.<sup>10,14</sup> The placebo (normal saline) was administered with an identical volume and according to the same bolus and infusion procedures as tirofiban. After near-completion of the study-drug infusion, subsequent antithrombotic therapy and other stroke care were provided according to contemporary guideline-based management.<sup>15</sup> For antiplatelet medications, their use was prohibited within 20 h post-enrolment. After 20 h post-medication, oral antiplatelet treatment could be initiated to ensure continuous platelet inhibition before stopping the study drug according to the consensus of experts in China.<sup>14</sup> Details of concomitant treatments are summarised in the protocol.

Baseline data included demographics, medical history, medication use, examination findings, stroke characteristics, and NIHSS scores. Follow-up assessments were done at 24 h (±12 h), 48 h (±12 h), day 7 (±1 day) or discharge (whichever occurred first), and day 90 (±7 days) after randomisation (appendix 1 p 31). To minimise loss to follow-up, at least three contact

modalities (such as personal mobile, family landline, or secondary relative's contact information) were obtained from each participant.

The mRS scores were assessed at 90 days and adjudicated by two independent neurologists who were certified in the use of the mRS and unaware of the treatment assignments. The assessments were based on video or audio recordings obtained during outpatient clinic visits, via telephone or video calls, or recorded by family members guided by the neurologists when live interaction was not feasible.<sup>16</sup> The NIHSS score (range 0–42, with higher scores indicating more severe neurological deficits) was assessed at 24 h, 48 h, and 7 days or at discharge by two independent, certified neurologists by video in a blinded manner, or, when video recordings were unavailable, collected by site investigators. Health-related quality of life was assessed at 90 days by the EQ-5D-5L questionnaire (ranging from –0.39 to 1.00, with higher scores indicating a better quality of life). Safety data were collected by site investigators at 24 h, 48 h, 7 days or at discharge, and 90 days to assess intracranial haemorrhage within 48 h and all-cause mortality within 90 days. Data from the source documentation were adjudicated by the imaging core laboratory and clinical event committee, who were blinded to treatment allocation.

### Outcomes

The primary efficacy outcome was functional independence, defined as a score of 0–2 on the mRS at 90 days after randomisation.

Secondary efficacy outcomes were the proportion of patients with excellent functional outcome (mRS score of 0–1) at 90 days; the proportion of patients with independent ambulation (mRS score of 0–3) at 90 days; the distribution of mRS score at 90 days; the proportion of patients with early neurological improvement at 36 h ( $\pm 12$  h), defined as an NIHSS score of 0–1 or a reduction of at least 4 points from baseline; and the EQ-5D-5L score at 90 days. All efficacy outcomes were assessed by the clinical event committee.

Safety outcomes were the proportion of patients with symptomatic intracranial haemorrhage (assessed according to the Heidelberg Bleeding Classification with “definite” or “probable” categories considered positive)<sup>17</sup> within 48 h after randomisation, the proportion of patients with any evidence of intracranial haemorrhage on imaging within 48 h, all-cause mortality within 90 days, and investigator-reported adverse events or serious adverse events.

### Statistical analysis

For the sample-size calculation, we assumed that 44.0% of patients in the placebo group would reach functional independence at 90 days.<sup>18</sup> Despite previous studies showing an absolute increase in the proportion of patients achieving functional independence of 10.3–22.6 percentage points with tirofiban after EVT, we

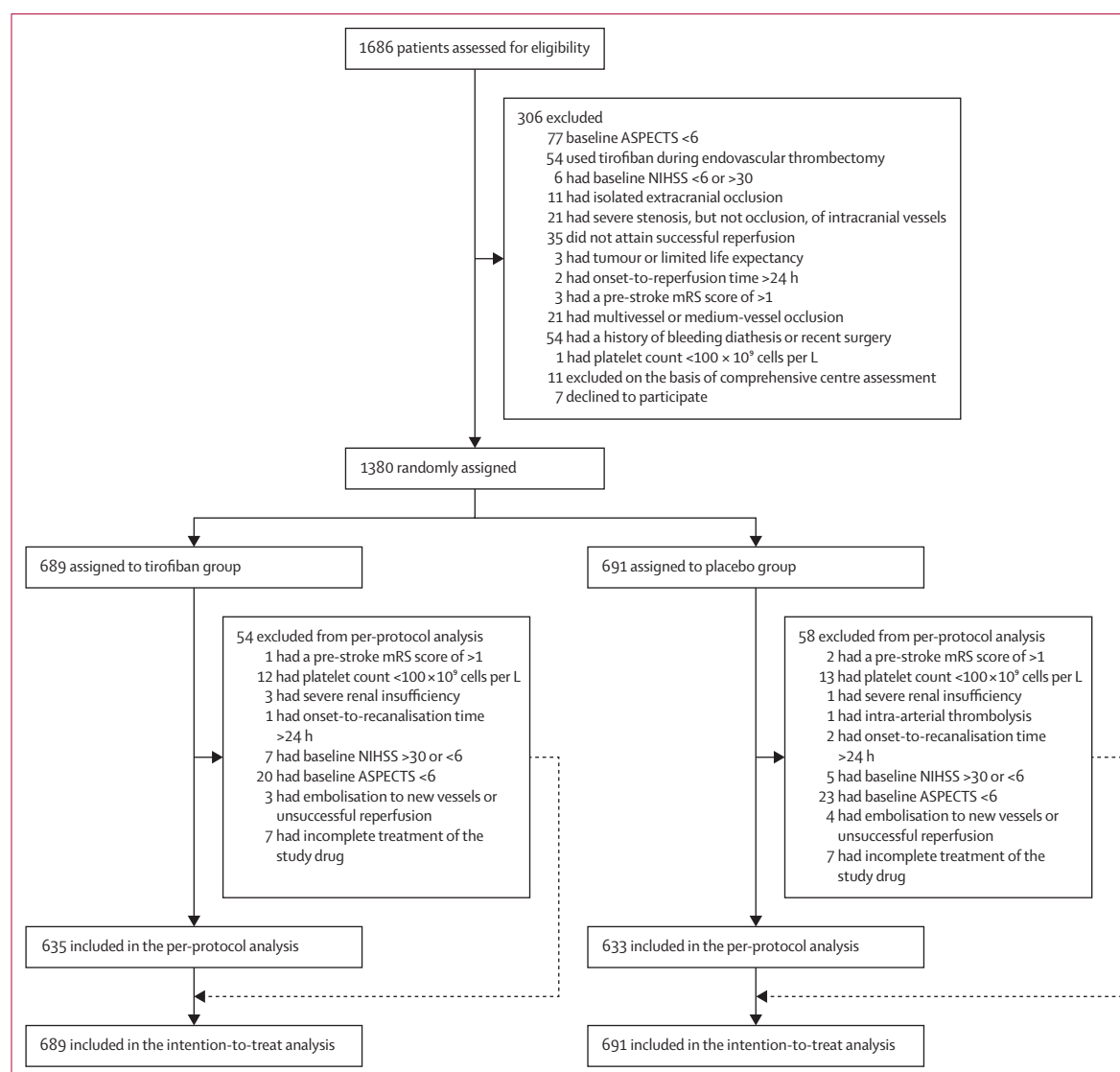
conservatively assumed an 8-percentage-point treatment effect.<sup>9,10,18</sup> With a two-sided alpha of 0.05, 80% power, and 10% loss to follow-up, 1360 patients (680 per group) were required. No interim efficacy analyses were done. An independent data and safety monitoring board periodically assessed safety throughout the trial.

Eligibility was assessed before randomisation on the basis of the information available at enrolment. Selected eligibility variables, including pre-stroke mRS and NIHSS, were subsequently reviewed during central data verification. Discrepancies between the initial eligibility assessment and the final central review were considered eligibility deviations if identified after randomisation. According to the prespecified intention-to-treat principle (ITT), all randomly assigned patients were retained in the ITT population. Patients with major eligibility deviations identified after randomisation were excluded from the per-protocol population.

The primary analyses followed the ITT principle and included all randomly assigned patients. Two sensitivity analyses were conducted: (1) in the ITT population with adjustment for centre effect and (2) in the per-protocol population. Safety analyses included all patients who received study treatment and had at least one safety assessment.

Between-group differences in the primary outcome were analysed using a modified Poisson regression model with robust variance estimation to estimate risk ratios (RRs) and corresponding 95% CIs. Both unadjusted and adjusted analyses were performed, with adjustment for prespecified baseline covariates including age, baseline NIHSS score, baseline ASPECTS, occlusion site, and time from last known well to randomisation. Binary secondary efficacy and safety outcomes were analysed using the same approach. The ordinal distribution of mRS scores at 90 days was analysed with ordinal logistic regression after verification of the proportional-odds assumption by the Brant test, with results expressed as a common odds ratio (OR; mRS scores 5–6 were combined to avoid counting a shift from death [mRS score of 6] to severe disability requiring constant nursing care [mRS score of 5] as a treatment success). The EQ-5D-5L scores were analysed with the win ratio method.<sup>19</sup> Mortality was analysed using Cox proportional hazards regression, with Kaplan–Meier curves constructed to display survival distributions and between-group comparisons performed using the log-rank test. For the Cox proportional hazards regression analysis of mortality, the proportional hazards assumption was assessed using the Schoenfeld residual test. As a sensitivity analysis to account for potential non-proportional hazards, we also estimated the restricted mean survival time (RMST) up to 90 days, with between-group differences calculated using the delta method.

Prespecified subgroup analyses were performed according to age (<71 years vs  $\geq 71$  years), sex (male vs female), time from last known well to randomisation



**Figure 1: Trial profile**

ASPECTS=Alberta stroke program early CT score. mRS=modified Rankin Scale. NIHSS=National Institutes of Health Stroke Scale.

(<6 h vs ≥6 h), baseline stroke severity (NIHSS score <16 vs ≥16), prestroke mRS score (0 vs 1), occlusion site (intracranial internal carotid artery vs middle cerebral artery M1/M2 segments), post-procedure mTICI grade (2b vs 3), stroke aetiology (large-artery atherosclerosis vs cardioembolism vs unknown or other), receipt of intravenous thrombolysis (yes vs no), and use of intraprocedural rescue therapy including angioplasty or stent placement (yes vs no).

Because no adjustment for multiplicity was performed for secondary or subgroup analyses, these results should be interpreted as exploratory. No imputation for missing data was performed because the data for the primary outcome and prespecified covariates were complete. Statistical analyses were performed with SAS software, version 9.4, and R software, version 4.3.0.

## Role of the funding source

This was an investigator-initiated trial. The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

## Results

From April 9, 2024, to Sept 29, 2025, a total of 1686 patients at 82 centres were assessed for eligibility, of whom 1380 were randomly assigned to either the tirofiban group (n=689) or the placebo group (n=691; figure 1; appendix 1 pp 21–23). No patients were lost to follow-up at 90 days; outcomes were obtained by telephone or audio assessment for 827 patients (408 [49%] from the tirofiban group and 419 [51%] from the placebo group) and by video assessment for



|                                                                                     | Tirofiban (n=689) | Placebo (n=691) |
|-------------------------------------------------------------------------------------|-------------------|-----------------|
| Age (years)                                                                         | 71 (63–77)        | 71 (62–77)      |
| Sex                                                                                 |                   |                 |
| Female                                                                              | 295 (43%)         | 296 (43%)       |
| Male                                                                                | 394 (57%)         | 395 (57%)       |
| Ethnicity                                                                           |                   |                 |
| Han Chinese ethnic group                                                            | 682 (99%)         | 685 (99%)       |
| Other ethnic groups (Tujia, Miao, Dong, and Daur)                                   | 7 (1%)            | 6 (1%)          |
| Median NIHSS score*                                                                 | 15 (12–19)        | 15 (11–19)      |
| Median ASPECTS†                                                                     | 8 (7–9)           | 8 (7–9)         |
| Medical history                                                                     |                   |                 |
| Hypertension                                                                        | 411 (60%)         | 407 (59%)       |
| Hyperlipidaemia                                                                     | 229 (33%)         | 217 (31%)       |
| Coronary heart disease                                                              | 150 (22%)         | 131 (19%)       |
| Diabetes                                                                            | 128 (19%)         | 130 (19%)       |
| Previous stroke or transient ischaemic attack                                       | 123 (18%)         | 133 (19%)       |
| Atrial fibrillation                                                                 | 329 (48%)         | 345 (50%)       |
| Smoking status                                                                      |                   |                 |
| Current smoker                                                                      | 148 (21%)         | 155 (22%)       |
| Former smoker                                                                       | 49 (7%)           | 40 (6%)         |
| Stroke aetiology‡                                                                   |                   |                 |
| Large-artery atherosclerosis                                                        | 202 (29%)         | 203 (29%)       |
| Cardioembolism                                                                      | 442 (64%)         | 443 (64%)       |
| Unknown and others                                                                  | 45 (7%)           | 45 (7%)         |
| Wake-up stroke                                                                      | 182 (26%)         | 189 (27%)       |
| Modified Rankin Scale score before stroke                                           |                   |                 |
| 0                                                                                   | 638 (93%)         | 633 (92%)       |
| 1                                                                                   | 50 (7%)           | 56 (8%)         |
| 2                                                                                   | 1 (<1%)           | 2 (<1%)         |
| Occlusion site                                                                      |                   |                 |
| Intracranial internal carotid artery                                                | 210 (30%)         | 203 (29%)       |
| M1 middle cerebral artery segment                                                   | 411 (60%)         | 423 (61%)       |
| M2 middle cerebral artery segment                                                   | 68 (10%)          | 65 (9%)         |
| Intravenous thrombolysis therapy                                                    | 240 (35%)         | 251 (36%)       |
| mTICI§                                                                              |                   |                 |
| 2b                                                                                  | 219 (32%)         | 225 (33%)       |
| 3                                                                                   | 470 (68%)         | 466 (67%)       |
| Interval between time last known well and puncture time (min)                       | 270 (190–437)     | 285 (188–444)   |
| Interval between time last known well and time of reperfusion (min)                 | 340 (245–507)     | 351 (243–510)   |
| Interval between time last known well and time of randomisation (min)               | 357 (259–526)     | 366 (254–520)   |
| Interval between time last known well and intra-arterial study agent infusion (min) | 360 (265–531)     | 370 (260–524)   |

Data are n (%) and median (IQR). Percentages may not total 100 because of rounding. ASPECTS=Alberta stroke program early CT score. mTICI=modified thrombolysis in cerebral infarction. NIHSS=National Institutes of Health Stroke Scale. \*Scores on the NIHSS range from 0 to 42, with higher scores indicating more severe neurological deficits. †ASPECTS is an imaging measure of the extent of ischaemic stroke; scores range from 0 to 10, with higher scores indicating a smaller infarct core. ‡The Trial of Org 10172 in Acute Stroke Treatment system classifies ischaemic stroke and transient ischaemic attack into five subtypes based on likely causes: large-artery atherosclerosis, cardioembolism, small-artery occlusion, other determined aetiology, and undetermined aetiology. §The mTICI scale measures reperfusion based on digital subtraction angiography (range from zero [no reperfusion] to three [complete reperfusion]).

Table 1: Baseline characteristics of the intention-to-treat population

553 patients (281 [51%] from the tirofiban group and 272 [49%] from the placebo group). Further, 112 (8%) patients with major protocol deviations were excluded from the per-protocol analysis (54 [8%] of 689 patients in the tirofiban group and 58 [8%] of 691 patients in the placebo group; figure 1).

Baseline characteristics were well balanced between the two groups in the ITT population (table 1, appendix 1 pp 33–34) and the per-protocol population (appendix 1 pp 35–36). The representativeness of the trial population is summarised in appendix 1 (p 32). The median age was 71 years (IQR 62–77), 591 (43%) patients were female and 789 (57%) were male, and 1367 (99%) were of Han Chinese ethnicity. The median NIHSS score was 15 (IQR 12–19), the median ASPECTS was 8 (IQR 7–9), and intravenous thrombolysis was administered to 491 (36%) patients. The cause of stroke was large-artery atherosclerosis in 405 (29%) patients, cardioembolism in 885 (64%) patients, and other or unknown in 90 (7%) patients.

For the primary efficacy analysis at 90 days, functional independence (mRS score of 0–2) was recorded in 340 (49%) patients in the tirofiban group and 299 (43%) patients in the placebo group (unadjusted absolute risk difference 6.1 percentage points, 95% CI 0.8–11.3,  $p=0.023$ ; adjusted absolute risk difference 6.5 percentage points, 95% CI 1.2–11.7,  $p=0.016$ ; adjusted RR 1.15, 95% CI 1.03–1.27,  $p=0.0092$ ; table 2, appendix 1 p 37). Results were consistent in sensitivity analyses, including models accounting for within-centre correlation using generalised estimating equations and analyses in the per-protocol population (appendix 1 p 43).

An excellent functional outcome (mRS score of 0–1) was recorded in 242 (35%) patients in the tirofiban group and 205 (30%) patients in the placebo group (adjusted RR 1.20, 95% CI 1.04–1.37,  $p=0.012$ ; table 2 and appendix 1 pp 37–40). Besides, the distribution of mRS scores significantly favoured tirofiban treatment (adjusted common OR 1.22, 95% CI 1.01–1.48;  $p=0.040$ ; figure 2). This favourable shift was corroborated in the per-protocol analysis (adjusted common OR 1.25, 1.02–1.52;  $p=0.031$ ; appendix 1 pp 41–42). Independent ambulation (mRS score of 0–3), early neurological improvement, and health-related quality of life (EQ-5D-5L) were similar between the groups (table 2). Results were also similar between the two groups in the per-protocol population (appendix 1 pp 41–42).

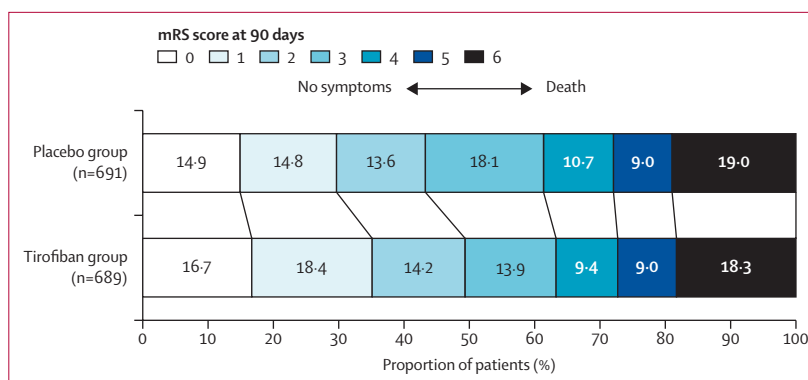
Regarding the safety analysis, symptomatic intracranial haemorrhage within 48 h occurred in 82 (12%) of 687 patients in the tirofiban group and 65 (9%) of 691 patients in the placebo group (adjusted RR 1.24, 95% CI 0.91–1.68;  $p=0.17$ ; table 2, appendix 1 p 44). Any intracranial haemorrhage within 48 h occurred in 235 (34%) patients in the tirofiban group and in 219 (32%) patients in the placebo group (adjusted RR 1.07, 95% CI 0.92–1.25;  $p=0.35$ ), with details of different haemorrhage subtypes provided in appendix 1 (p 46). 126 (18%) patients in the tirofiban group and

|                                                  | Tirofiban group (n=689) | Placebo group (n=691)   | Effect measure    | Adjusted effect value (95% CI)* | p value |
|--------------------------------------------------|-------------------------|-------------------------|-------------------|---------------------------------|---------|
| <b>Primary efficacy outcome</b>                  |                         |                         |                   |                                 |         |
| mRS score of 0–2 at 90 days†                     | 340 (49%)               | 299 (43%)               | Risk ratio        | 1.15 (1.03–1.27)                | 0.0092  |
| <b>Secondary efficacy outcomes</b>               |                         |                         |                   |                                 |         |
| mRS score of 0–1 at 90 days                      | 242 (35%)               | 205 (30%)               | Risk ratio        | 1.20 (1.04–1.37)                | 0.012   |
| mRS score of 0–3 at 90 days                      | 436 (63%)               | 424 (61%)               | Risk ratio        | 1.04 (0.96–1.12)                | 0.37    |
| Ordinal distribution of mRS at 90 days‡          | 3 (1–5)                 | 3 (1–5)                 | Common odds ratio | 1.22 (1.01–1.48)                | 0.040   |
| Early neurological improvement§                  | 426 (62%)               | 427 (62%)               | Risk ratio        | 1.01 (0.93–1.09)                | 0.89    |
| EQ-5D-5L score at 90 days                        |                         |                         |                   |                                 |         |
| Median value (IQR)                               | 0.59 (0.06–0.96)        | 0.65 (0.06–0.96)        | ..                | ..                              | ..      |
| Number of wins/total number of pairs¶            | 217 299/476 099 (45.6%) | 212 222/476 099 (44.6%) | Win ratio¶        | 1.04 (0.91–1.18)                | 0.61    |
| <b>Safety outcomes</b>                           |                         |                         |                   |                                 |         |
| Symptomatic intracranial haemorrhage within 48 h | 82/687 (12%)            | 65/691 (9%)             | Risk ratio        | 1.24 (0.91–1.68)                | 0.17    |
| Any intracranial haemorrhage within 48 h         | 235/687 (34%)           | 219/691 (32%)           | Risk ratio        | 1.07 (0.92–1.25)                | 0.35    |
| All-cause mortality within 90 days               | 126 (18%)               | 131 (19%)               | Hazard ratio      | 0.96 (0.75–1.22)                | 0.72    |

Data are n (%), number of wins/total number of pairs, or n/N (%), unless otherwise stated. ASPECTS=Alberta stroke program early CT score. mRS=modified Rankin Scale. NIHSS=National Institutes of Health Stroke Scale. \*Adjusted for age, baseline NIHSS score, baseline ASPECTS score, occlusion site, and time from last known well to randomisation, but were not adjusted for multiple comparisons. †The mRS of functional disability ranges from 0 (no symptoms) to 6 (death). ‡mRS scores 5–6 were combined to avoid counting a shift from death (mRS score of 6) to severe disability requiring constant nursing care (mRS score of 5) as a treatment success. §Early neurological improvement was defined as an NIHSS score of 0 to 1 or a reduction of at least 4 points at 36 h (±12 h) from baseline. ¶For the win ratio analysis, a win was defined as a better health status across the five EQ-5D dimensions, compared hierarchically. If patients were tied on one dimension, subsequent dimensions were compared sequentially. A win ratio greater than 1 indicates a favourable outcome in the tirofiban group. ||Two patients in the tirofiban group were not evaluable for intracranial haemorrhage due to absence of post-procedural imaging.

**Table 2: Primary and secondary outcomes**

131 (19%) patients in the placebo group died within 90 days after randomisation (adjusted hazard ratio 0.96, 95% CI 0.75–1.22;  $p=0.72$ ; table 2). The Schoenfeld residual test suggested evidence of a potential departure from proportional hazards for the mortality analysis ( $p=0.013$ ); however, visual inspection of the log(–log[survival]) curves did not suggest a clinically meaningful time-varying treatment effect (appendix 1 pp 26, 28). In a sensitivity analysis that did not rely on the proportional hazards assumption, the RMST up to 90 days was 76.1 days in the tirofiban group and 76.9 days in the placebo group (difference  $-0.77$  days, 95% CI  $-3.88$  to  $2.33$ ;  $p=0.63$ ), consistent with the Cox regression results. Results were similar in the per-protocol population (appendix 1 p 45). Kaplan–Meier estimates of survival are shown in appendix 1 (pp 25, 27). The incidence of serious adverse events and any adverse events was similar between the two groups. A detailed summary of adverse events is provided in appendix 1 (p 47). Serious adverse events occurred in 275 (40%) of 689 patients in the tirofiban group and 280 (41%) of 691 patients in the placebo group (table 3). The most common serious adverse event categories were infections and infestations (152 [22%] patients in the tirofiban group vs 176 [25%] patients in the placebo group), nervous system disorders (121 [18%] patients in the tirofiban group vs 105 [15%] patients in the placebo group), and cardiac disorders (43 [6%] patients in the tirofiban group vs 60 [9%] patients in the placebo group).



**Figure 2: Distribution of mRS scores at 90 days in the intention-to-treat population**

Distribution of scores on the mRS among patients in the tirofiban group and the placebo group. The mRS is a 7-point ordinal scale of functional disability. Scores range from 0 to 6, with 0 indicating no symptoms, 1 no clinically significant disability, 2 slight disability, 3 moderate disability, 4 moderately severe disability, 5 severe disability, and 6 death. No patients were lost to follow-up at 90 days. Percentages might not total 100 because of rounding. mRS=modified Rankin Scale.

In the prespecified subgroup analyses of the primary outcome, the direction of the treatment effect consistently favoured tirofiban across major clinical and procedural subgroups in the ITT and per-protocol populations (figure 3, appendix 1 p 29). There was no significant evidence of heterogeneity, and all interaction tests were non-significant. Although point estimates varied across subgroups, no clinically meaningful deviation from the overall treatment effect was observed.

|                                                      | Tirofiban<br>(n=689) | Placebo<br>(n=691) |
|------------------------------------------------------|----------------------|--------------------|
| Overall*                                             | 275 (40%)            | 280 (41%)          |
| Blood and lymphatic system disorders                 | 13 (2%)              | 7 (1%)             |
| Cardiac disorders                                    | 43 (6%)              | 60 (9%)            |
| Gastrointestinal disorders                           | 12 (2%)              | 9 (1%)             |
| General disorders and administration site conditions | 2 (<1%)              | 1 (<1%)            |
| Hepatobiliary disorders                              | 4 (1%)               | 1 (<1%)            |
| Infections and infestations                          | 152 (22%)            | 176 (25%)          |
| Metabolism and nutrition disorders                   | 6 (1%)               | 11 (2%)            |
| Musculoskeletal and connective tissue disorders      | 0                    | 1 (<1%)            |
| Neoplasms (benign, malignant, and unspecified)       | 3 (<1%)              | 3 (<1%)            |
| Nervous system disorders                             | 121 (18%)            | 105 (15%)          |
| Renal and urinary disorders                          | 3 (<1%)              | 2 (<1%)            |
| Respiratory, thoracic, and mediastinal disorders     | 26 (4%)              | 25 (4%)            |
| Skin and subcutaneous tissue disorders               | 1 (<1%)              | 1 (<1%)            |
| Surgical and medical procedures                      | 6 (1%)               | 4 (1%)             |
| Vascular disorders                                   | 16 (2%)              | 13 (2%)            |
| Endocrine disorders                                  | 1 (<1%)              | 0                  |

Data are n (%). Serious adverse events were classified using the fifth edition of the Common Terminology Criteria for Adverse Events of the USA. \*Overall indicates the number of patients with at least one serious adverse event. For each system organ class, patients with multiple events of one type were counted once. Patients could have serious adverse events in more than one system organ class; therefore, the numbers across system organ classes should not be summed.

**Table 3: Serious adverse events**

In post-hoc exploratory analyses stratified by stroke aetiology, among patients with large-artery atherosclerosis, tirofiban was associated with higher proportions of functional independence at 90 days (RR 1.29, 95% CI 1.08–1.53) without an increase in symptomatic intracranial haemorrhage (0.81, 0.38–1.74; appendix 1 p 48). By contrast, among patients with cardioembolic stroke, no clear functional benefit was observed for functional independence (1.08, 0.94–1.23), and, although not significant, the point estimate for symptomatic intracranial haemorrhage pointed towards an increased risk (RR 1.41, 0.98–2.01; appendix 1 p 49).

In post-hoc exploratory analyses stratified by intravenous thrombolysis, the point estimate for functional independence at 90 days favoured tirofiban both in patients who did not receive intravenous thrombolysis (adjusted RR 1.21, 95% CI 1.06–1.38) and in those who received intravenous thrombolysis (1.08, 0.92–1.28; figure 3). The risk of symptomatic intracranial haemorrhage was not clearly increased in either subgroup (p for interaction=0.71): among patients without intravenous thrombolysis, symptomatic intracranial haemorrhage occurred in 53 (12%) of 447 patients in the tirofiban group and 38 (9%) of 440 in the placebo group (adjusted RR 1.29, 0.85–1.96), whereas among

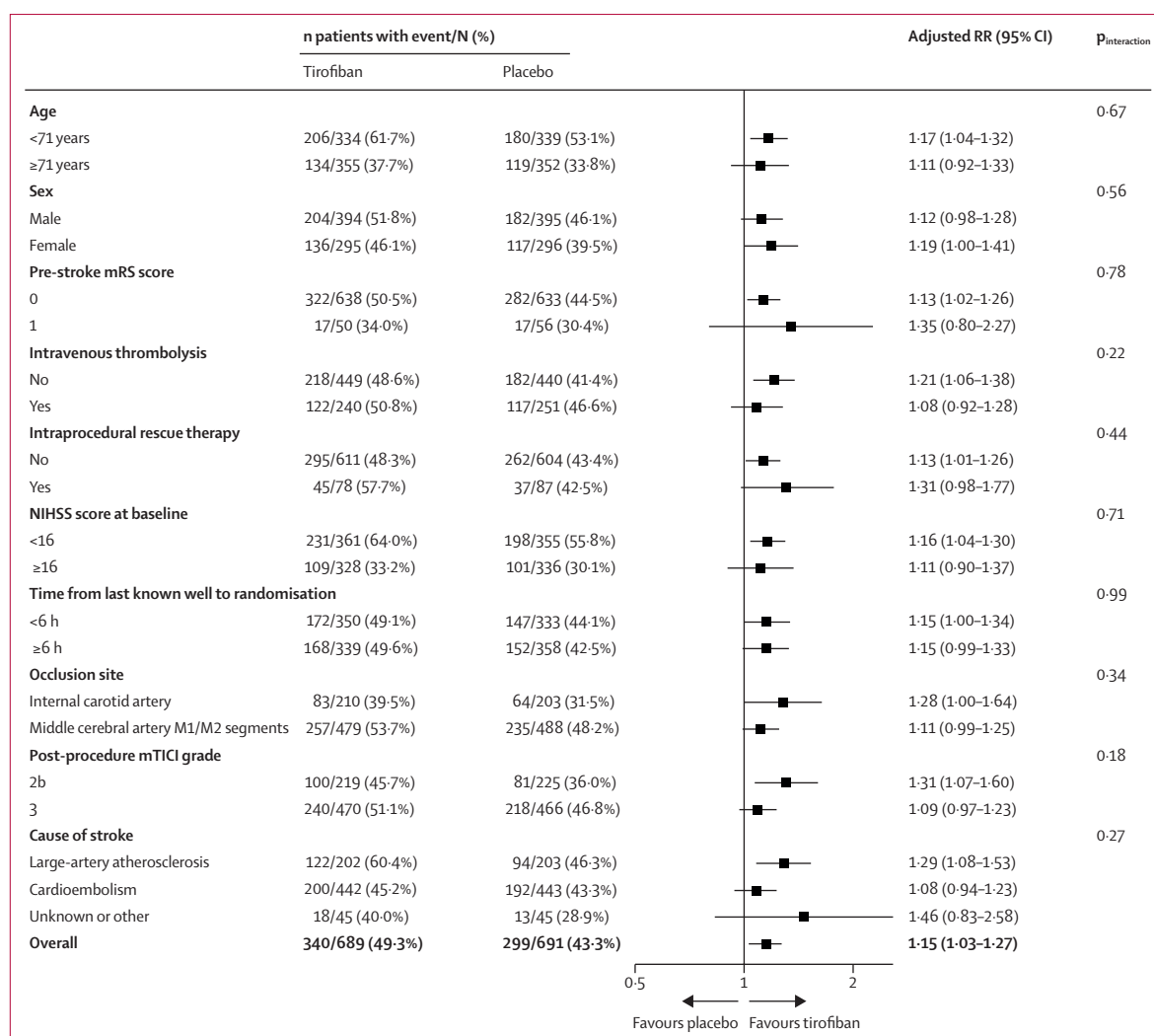
patients who received intravenous thrombolysis, the corresponding proportions were 29 (12%) of 240 patients in the tirofiban group and 27 (11%) of 251 patients in the placebo group (1.13, 0.65–1.95; appendix 1 p 30). In post-hoc exploratory analyses stratified by time from last known well to randomisation, no significant interaction was observed across time categories (p for interaction=0.34 for five categories and 0.85 for three broader categories; appendix 1 p 50). Because relatively few patients were randomly assigned in later time windows (1054 [76%] patients were randomly assigned within 9 h), these analyses were underpowered to assess treatment effects beyond 9 h and should be interpreted cautiously. In a post-hoc exploratory subgroup analysis according to previous antiplatelet therapy, the point estimate for functional independence favoured tirofiban among patients without previous antiplatelet therapy (adjusted RR 1.16; 95% CI 1.04–1.29), whereas the estimate was closer to the null among patients with previous antiplatelet therapy (1.02; 0.66–1.58; appendix 1 p 55). However, there was no significant interaction between previous antiplatelet therapy and treatment effect (p for interaction=0.63). These post-hoc findings should be interpreted as exploratory and hypothesis-generating, rather than as evidence of a definitive differential treatment effect according to subgroup status.

The trial eligibility criteria required an ASPECTS of 6 or higher, and only a small number of patients with ASPECTS lower than 6 were randomly assigned. In post-hoc exploratory analyses according to ASPECTS categories, the effect of tirofiban on functional independence at 90 days did not differ significantly across patients with ASPECTS lower than 6, ASPECTS 6–7, and ASPECTS 8–10 (p for interaction=0.55; appendix 1 pp 52–54). The adjusted RR for mRS 0–2 at 90 days was 0.94 (95% CI 0.51–1.73) among patients with ASPECTS lower than 6, 1.23 (0.95–1.59) among those with ASPECTS 6–7, and 1.15 (1.02–1.28) among those with ASPECTS 8–10. There was also no significant heterogeneity by ASPECTS for symptomatic intracranial haemorrhage (p for interaction=0.78), with adjusted RRs of 1.14 (0.40–3.26) among those with ASPECTS lower than 6, 1.12 (0.68–1.82) among those with ASPECTS 6–7, and 1.36 (0.89–2.06) among those with ASPECTS 8–10.

## Discussion

In this multicentre, randomised, double-blind trial involving patients with anterior-circulation large-vessel occlusion achieving successful reperfusion after endovascular thrombectomy, adjunct tirofiban administered as an intra-arterial bolus followed by an intravenous infusion led to a higher rate of patients with functional independence at 90 days than did placebo (49% vs 43%; adjusted RR 1.15, 95% CI 1.03–1.27). The incidence of symptomatic intracranial haemorrhage and death was not significantly different between the groups. Although the between-group difference was not significant, symptomatic





**Figure 3: Subgroup analyses of the primary outcome in the intention-to-treat population**

Forest plot comparing the tirofiban group with the placebo group according to subgroup with respect to the finding of a score of 0 to 2 on the mRS at 90 days (the primary outcome). The trial was not powered and had no prespecified correction for multiple comparisons for a definitive analysis of subgroups. The RR was adjusted for age, baseline NIHSS score, baseline ASPECTS, occlusion site, and time from last known well to randomisation. The median age of the patients was 71 years. Subgroups are defined as younger than 71 years and as age 71 years or older. Scores on the NIHSS range from 0 to 42, with higher scores indicating greater neurological deficits; a score of 16 or higher on this scale indicates a severe stroke. ASPECTS=Alberta stroke program early CT score. mRS=modified Rankin Scale. mTICI=modified thrombolysis in cerebral infarction. NIHSS=National Institutes of Health Stroke Scale. RR=risk ratio.

intracranial haemorrhage occurred numerically more often with tirofiban than with placebo. In exploratory subgroup analyses, the safety profile appeared less favourable among patients with cardioembolic stroke, although these findings were not definitive and should be interpreted cautiously because the trial was not powered for subgroup-specific safety comparisons. Supporting these efficacy signals, the proportion of patients with excellent functional outcome (mRS 0–1) at 90 days and ordinal mRS shift analysis also seemed to favour tirofiban, whereas early neurological improvement and EQ-5D-5L scores were similar between the two groups.

The findings of this trial should be interpreted in the context of previous studies of adjunctive antithrombotic

therapy in endovascular thrombectomy. In the RESCUE BT trial, intravenous tirofiban did not improve functional outcomes and was associated with increased intracranial haemorrhage (34.9% with tirofiban vs 28.0% with placebo), and no significant difference in symptomatic intracranial haemorrhage (9.7% vs 6.4%).<sup>20</sup> Unlike our trial, tirofiban in the RESCUE BT trial was initiated before thrombectomy, and it was administered at a higher maintenance dose (0.15 µg/kg per min) than the dose used in our trial, regardless of the reperfusion status. These differences suggest that the effects of tirofiban might depend not only on the agent itself, but also on the timing of initiation and the physiological context in which platelet inhibition is introduced. Similarly, in the MR

CLEAN-MED trial, periprocedural aspirin or heparin (initiated either immediately after groin puncture or, if intravenous thrombolysis was ongoing, upon its completion) increased symptomatic intracranial haemorrhage without improving functional outcomes.<sup>21</sup> Compared with MR CLEAN-MED, ATTRACTION differed in the antithrombotic agent (a glycoprotein IIb/IIIa inhibitor rather than aspirin or heparin), and the timing of initiation (after achieving reperfusion rather than immediately after puncture), while restricting enrolment to patients with confirmed successful reperfusion. This design isolates a phase of stroke pathophysiology in which macrovascular recanalisation has been attained but microvascular thrombosis and reperfusion injury might still be evolving, and it could help to explain the different balance of efficacy and safety observed.

Prespecified subgroup analyses showed a consistent direction of treatment effect across major clinical and procedural subgroups, with no significant evidence of heterogeneity. In exploratory analyses, the apparent balance between benefit and risk appeared to vary numerically by stroke aetiology, although the test for interaction was not significant. Patients with large-artery atherosclerosis seemed to show a larger point estimate for functional benefit without an apparent increase in haemorrhagic complications, whereas in cardioembolic stroke, no clear functional benefit of tirofiban was observed and point estimates suggested an increased risk of intracranial haemorrhage. A similar pattern has been reported in post-hoc analyses of the RESCUE BT trial, with a suggestive benefit of tirofiban in patients with large-artery atherosclerotic stroke, yet poorer outcomes and potential safety concerns in patients with cardioembolic stroke.<sup>18,22</sup> These findings are biologically plausible, as large-artery atherosclerotic stroke is more commonly associated with residual stenosis and platelet-rich in-situ thrombosis after thrombectomy.<sup>23,24</sup> They raise the possibility that stroke aetiology could influence the benefit–risk balance of tirofiban administered after successful reperfusion, but this hypothesis warrants validation in future studies formally powered to evaluate treatment-effect heterogeneity. The ongoing ANGEL-DRUG trial (NCT07026318), which enrolls only patients with 50% or higher residual stenosis and excludes those with a confirmed cardioembolic source, could help further clarify whether the benefit–risk profile of tirofiban varies by underlying stroke aetiology. In addition, the apparent treatment effect in this trial was less evident among patients who received intravenous thrombolysis than among those who did not. This pattern might be biologically plausible, because previous thrombolytic therapy could influence residual thrombus burden, platelet-mediated microvascular thrombosis, and haemorrhagic risk after reperfusion. It may also have improved outcomes in the placebo group, thereby reducing the between-group difference. However, given the absence of a significant interaction, these findings should be

interpreted as exploratory and hypothesis-generating rather than as evidence of a definitive differential treatment effect according to stroke aetiology or previous intravenous thrombolysis.

Intracranial haemorrhage is one of the most frequent adverse events following endovascular treatment, and symptomatic intracranial haemorrhage is the most clinically concerning form of haemorrhagic complication given its strong association with poor prognosis. In the present trial, symptomatic intracranial haemorrhage was numerically more common in the tirofiban group than in the placebo group, although the between-group difference was not significant. A systematic review of tirofiban in patients undergoing mechanical thrombectomy has reported inconsistent findings regarding bleeding and mortality across different study protocols, with no consistent signal of increased symptomatic intracranial haemorrhage.<sup>25</sup> In the review, rates of symptomatic intracranial haemorrhage ranged from 2.9% to 22.2%. The overall rate of 12% in the tirofiban group observed in our study fell within this previously reported range. Although the difference in symptomatic intracranial haemorrhage was not significant, the numerical excess with tirofiban and the exploratory subgroup signal among patients with cardioembolic stroke warrant cautious interpretation. Moreover, 90-day mortality was similar between groups, as was the overall burden of adverse events.

Several randomised trials have evaluated adjunctive intra-arterial thrombolysis after successful or near-complete endovascular reperfusion as a strategy to improve tissue-level reperfusion. While individual trial results remain heterogeneous (appendix 1 p 51),<sup>26–29</sup> a recent meta-analysis indicated that this approach might improve functional outcomes without a concomitant increase in symptomatic intracranial haemorrhage.<sup>30</sup> Our study evaluated a distinct pharmacological strategy. Unlike intra-arterial fibrinolytics, which primarily target residual distal emboli and fibrin-rich microthrombi, tirofiban inhibits glycoprotein IIb/IIIa-mediated platelet aggregation, thereby potentially mitigating platelet-driven microvascular thrombosis or acute re-occlusion. Intriguingly, our post-hoc analysis revealed a potential divergence in treatment response across aetiological subgroups. Whereas adjunctive thrombolysis has historically demonstrated a more pronounced benefit in cardioembolic stroke, our data suggest a preferential advantage for tirofiban in patients with large-artery atherosclerosis. While these findings are hypothesis-generating and preclude definitive causal inference, they provide a signal that the efficacy of adjunctive pharmacotherapy might be aetiology-specific, reflecting the distinct pathophysiological substrates of the microvasculature post-recanalisation. Taken together, these findings bolster the concept that post-procedural pharmacological therapy can enhance functional recovery. Nevertheless, the optimal therapeutic regimen remains

unestablished, necessitating further investigation to determine if platelet inhibition, fibrinolysis, or a combination thereof provides superior clinical utility.

The absence of significant between-group differences in early neurological improvement and EQ-5D-5L might reflect the distinct timeframes and constructs captured by these outcomes. Early neurological improvement might be less sensitive to post-reperfusion tirofiban if its effects are mediated through prevention of re-occlusion,<sup>25</sup> which might contribute more to later recovery than to neurological change in the early post-stroke period. Additionally, EQ-5D-5L assesses health-related quality of life across multiple domains, including pain, emotional wellbeing, and usual activities, and it is therefore not equivalent to functional status as measured by the mRS. Therefore, the absence of significant between-group differences in these measures does not necessarily contradict the observed benefit in functional independence, although it supports a cautious interpretation of the overall efficacy profile.

ASPECTS is an important marker of baseline infarct extent and may plausibly influence both the potential benefit and bleeding risk of adjunctive antiplatelet therapy after endovascular reperfusion. In post-hoc exploratory analyses, we did not observe significant evidence that ASPECTS modified the effect of tirofiban on functional independence or intracranial haemorrhage outcomes. However, the trial was designed primarily to enrol patients with ASPECTS 6–10, and only a small number of patients had ASPECTS lower than 6, resulting in imprecise estimates in this subgroup. Although the point estimate in patients with ASPECTS lower than 6 did not suggest a favourable effect of tirofiban, this finding should be interpreted cautiously and should not be taken as definitive evidence of absence of benefit or harm. Therefore, the present findings are most applicable to patients with ASPECTS 6–10 after successful endovascular reperfusion, whereas their generalisability to patients with lower ASPECTS remains uncertain.

To the best of our knowledge, ATTRACTION is a large-scale, multicentre, double-blind randomised trial with complete follow-up and careful ascertainment of clinical outcomes. By enrolling only patients with more than 50% reperfusion of the affected brain territory (mTICI 2b–3), the trial specifically addressed a clinically relevant but previously underexplored therapeutic window. The pragmatic intervention, an intra-arterial bolus followed by a short intravenous infusion, could potentially facilitate implementation in routine clinical practice.

Nevertheless, several limitations should be acknowledged. First, the study was conducted exclusively in China, hence external validation in other populations is warranted. Second, the intervention combined an intra-arterial bolus with intravenous infusion, and the relative contributions of each component cannot be determined. Third, dual-energy CT was not routinely available across participating centres and was not mandated in the

protocol. Therefore, although the independent imaging core laboratory considered the potential distinction between contrast staining or contrast extravasation and haemorrhagic transformation, some overestimation of radiological haemorrhage remains possible. Fourth, the generalisability of the findings should be interpreted in light of the trial eligibility criteria and the distribution of enrolled patients. The results might not be generalisable to patients with less than 50% reperfusion of the affected brain tissue or those with lower ASPECTS. Fifth, although enrolment was permitted up to 24 h after last known well, most patients were treated relatively early, and the evidence for very late treatment remains limited. Sixth, subgroup analyses according to previous intravenous thrombolysis and stroke aetiology were exploratory and not powered to establish definitive treatment effects within individual subgroups; these findings therefore warrant further investigation. Finally, our findings should be interpreted as applying to patients with mTICI 2b–3 reperfusion and should not be extrapolated exclusively to patients with near-complete or complete reperfusion.

In conclusion, in patients with anterior-circulation large-vessel occlusion who had successful reperfusion after thrombectomy, adjunctive tirofiban was associated with an increased rate of functional independence at 90 days. Although there was no significant overall increase in symptomatic intracranial haemorrhage, the numerical excess and exploratory findings in the cardioembolic stroke subgroup underscore bleeding risk as an important consideration when administering tirofiban to patients with acute ischaemic stroke after successful reperfusion. These findings suggest that tirofiban might represent a promising adjunctive therapeutic option after successful endovascular reperfusion, but caution is warranted regarding the risk of haemorrhage, especially in cardioembolic stroke. Nonetheless, further studies are needed to validate these results in other ethnic cohorts, and to explore whether the efficacy and safety of post-reperfusion tirofiban differ across specific stroke aetiological subtypes, particularly large-artery atherosclerosis and cardioembolic stroke.

#### Contributors

XL, WW, and ZT proposed the study conception and study design. HH, SX, and MW supervised the study design. HH, SX, TQ, and MW prepared the first draft of the report. MG, TNN, DW, and JS provided essential revisions, and all coauthors reviewed the article. HH, CL, GD, YW, DA, YX, YZ, QK, LL, TY, WC, WZ, JLS, PZ, FL, BM, PJ, JZ, and JD were involved in trial conduct, patient management, and data collection. YY, SH, and ZY supported data management. XL, HH, TQ, GL, and DW prepared the statistical analysis plan. TQ and DW conducted statistical analyses. XL, HH, SX, TQ, and MW accessed and verified the underlying data reported in the manuscript. All authors had final responsibility for the decision to submit for publication.

#### Declaration of interests

TNN holds a compensated editorial position (Associate Editor) at *Stroke* and reports receiving personal consulting fees from Bayer, Route 92, and Medtronic. JLS serves as an Associate Editor at *JAMA* and reports having received personal fees from Roche, Genentech, and Medtronic and stock options from Rapid Medical. MG reports having received grants or contracts from Medtronic and Cerenovus; royalty payments from

Microvention; consulting fees from Microvention Penumbra, Philips, Stryker, and Medtronic Medtronic; and stock options from Circle Neurovascular. All other authors declare no competing interests. No confidentiality agreements existed between the manufacturer and investigators, and the manufacturer had no authority to delay or restrict publication of trial findings.

#### Data sharing

De-identified individual participant data, the study protocol, statistical analysis plan, and statistical code will be made available after publication of the primary results. Access will be granted to qualified researchers upon reasonable request to the corresponding author, following approval by the steering committee and completion of a data access agreement. Data will be shared for scientifically sound proposals and might involve collaboration with the study investigators.

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